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An efficient and provable certificate-based proxy signature scheme for IIoT environment

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ABSTRACT

Recently, the deployment of sensors and actuators to collect and disseminate data in various applications such as e-healthcare, vehicular adhoc networks (VANETs) and smart factories has revolutionized several new communication technologies. The Internet of Things (IoT) is one of those emerging communication technologies. These revolutionary applications of IoT in industrial environment are termed as Industry 4.0 and it has vitalized the concept of Industrial IoT (IIoT). Being wireless communication, the authentication and integrity of data are the most important challenges. To mitigate these challenges, several digital signature schemes are proposed in the literature. However, due to identity-based or certificate-less construction, those schemes suffer from inborn key escrow and secret key distribution problems. To resolve such issues, the first certificate-based proxy signature (PFCBPS) scheme without pairing is proposed. The proposed PFCBPS scheme is provably secure in random oracle model (ROM). The performance comparison (in terms of computational costs of different phases and length of resulting delegation and signature) shows that the proposed PFCBPS scheme's total computational cost is 46.69 ms which is 52.24% of He et al. [8], 61.40% of Debiao et al. [5], 23.33% of Seo et al. [20], 28% of Hu et al. [9] and 36.84% of Verma and Singh [23]. Thus, it is more suitable to IIoT environment than existing competitive schemes.

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1. Introduction

In last decade, the methods of collection and dissemination of data have inspired the concept of Internet of Things (IoT). In IoT, the sensors and actuators are deployed on smart vehicles, smart machines (in factories), remote location patients (implanted or wearable), etc. to collect and disseminate the data (Fig. 1). These smart vehicles can handle the odd traffic

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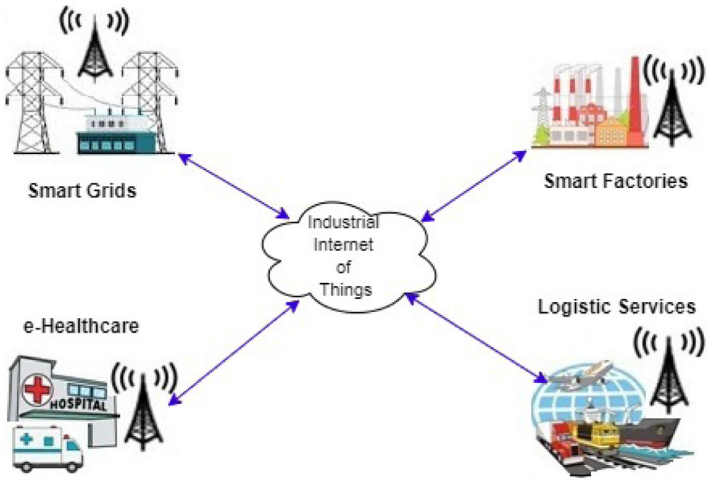


Fig. 1. Industrial IoT applications.

situations such as - congestion or traffic jam on road by sending the real time data to other vehicles. The sensors deployed on remote location patient can share the real time data of the patient such as pH-value, heartbeat rate, diabetes level, etc. to a medical professional via e-health cloud. In industrial applications, the smart machines can manage the temperature, electric intensity, number of process cycles (based on type of industry), production process, etc. and can output the optimized results. The combination of such IoT-based applications has emerged as Industrial IoT (IIoT). To create communication between such heterogeneous devices, the three tier infrastructure is required. The three tiers are classified as smart devices, Edge gateways and clouds. Thus devices send their data to a receiver through the gateway and cloud. The different standards used for connection and communication between the source and sender are IETF 6LoWPAN, IPv6, IETF's Constrained Application Protocol, ZeroMQ, and MQTT. The Fog computing and Edge computing are also used to improve the communication quality. The ease of collection and dissemination of data in IIoT has optimized the production cost, transportation cost and profit.

In the third industrial revolution (Industry 3.0) the automation of industrial processes and machines was emerged by combination of computational, electrical and electronic tools. The Printed Circuit Boards, cables and wires were few of them which were used in that era. Recently, the emerging deployment of IoT devices or concepts invented the notion of Industry 4.0 in the form of IIoT. In IIoT, cyber-physical systems (CPS) are deployed on industrial processes by replacing the programmable logic units. The CPS is an integration of cloud computing, big data, mobile computing, etc. and uses the wireless communication mostly. In this system, the machines and humans communicates to each other via wireless channels. The source sends the data to receiver via clouds. Being the wireless communication, the IIoT infrastructure is the most sensitive to a cyber attack. For example, in a smart logistic infrastructure, an attack may cause an accident and may result in a loss of human lives or resources. Thus, the authentication of data before storing to cloud is an important issue. The method of attestation can be used to authenticate the source and the data. The digital signature schemes are the method of such attestation mechanism.

In IIoT environment, the executive officer gives directives to their subordinates to perform the official communications. Therefore, the subordinate can handle the official messages communication on behalf of its officer. Thus, the authentication or attestation should be done by subordinate on behalf of its executive and the receiver must also be convinced. To implement such attestation mechanism, the delegation of signing power is required. Thanks to Mambo et al. [16], who proposed the method of delegation of signing power. They proposed this method in the form of proxy signature (PS) scheme. In PS scheme, two signers are involved, who are known as original signer and proxy signer. Original signer delegates its signing power to proxy signer and proxy signer can attest the data. In IIoT environment, executive officer acts as original signer and subordinate acts as proxy signer. Thus, subordinate uses the PS scheme to attest the data before sending.

The PS scheme by Mambo et al. [16], is designed by using the idea of public key cryptography (PKC) [6]. In PKC, every signer (proxy or original) generates its public and private key pair. Then, sends his/her public key to certification officer/authority (CO) to get certificates. In PKC, during the signing process, the certificates are used in explicit manner by the signer. Thus, the verifier is unable to check the validity period of the certificate. For example, signer joins a new company and uses the certificates issued from old company. In this scenario, the verifier cannot confirm the certificate's validity. Therefore, to check the validity, verifier requests the CO. The requests towards validity check of certificates need a huge infrastructure and it is known as third party queries (TPQ) or certificate status (CS) problem.

To mitigate TPQ/CS problem, the concept of identity-based PKC (IDBPKC) was proposed by Shamir [21]. In IDBPKC, a trusted officer/authority (TA) generates the private key of the signer in such a way that public key is derivable from the identity (ID) of the signer. Thus, no certification of public key is needed. Therefore, IDBPKC eliminates the CS or TPQ prob-

lems. However, TA knows the secret key. Thus, TA can use this in an illegal manner, i.e., to sign an unwanted message. This unwanted use of secret key is called key escrow (KE) problem. Besides this, IDBPKC faces another problem called secret key distribution (SKD) problem. The reason of SKD problem is the need of a secure channel for sending the secret key between TA and owner of the key.

To mitigate KE problem, the idea of certificate-less PKC (CLPKC) was proposed by Al-Riyami and Paterson [3]. In CLPKC, TA does not generate private key, but it generates partial private key. This partial key is sent to owner via a secure channel. After receiving this, owner integrates partial key and a self chosen secret value to generate private key. Now, TA is unable to get private key. Thus, CLPKC eliminates KE problem. However, key distribution problem is still available. Due to lack of certification, CLPKC suffers "public key replacement" attack. In this attack, adversary replaces the public key with a self chosen value and the verification process can be done with this fake public key.

Parallel to CLPKC, Gentry proposed the concept of certificate-based PKC (CBPKC) [7] to solve inherited key escrow. However,the merits of traditional PKC and IDBPKC are preserved. In this notion, user creates its key pair (private/public) as it does in PKC and then obtain a certification on identity (ID) and public key from CO. In CBPKC, certificates are implicitly used as a decryption (or signing) key and thus, each time an updated certificate is obtained by its owner. Therefore, CBPKC eliminates the secret key distribution problem and key escrow of IDBPKC. Based on the updation of certificates, CBPKC also solves the problems associated to certificate revocation.

1.1. Related work

In 2004, Kang et al. [11] proposed the first certificate-based digital signature (CBDS) scheme. The purpose of the proposal was to mitigate KE, TPQ and KDP problems from attestation method. However, in 2007, a key replacement attack on their scheme was proposed by Li et al. [14]. Parallel to the attack, the authors also proposed an improved threat model. In 2013, Li et al. [15] proposed a cryptanalysis of the CBDS scheme by Ming and Wang [17]. The scheme by Ming and Wang was proposed in pairing free settings and hence, it was very efficient. Therefore, to retain efficiency, Li et al. [15] also proposed an improved and secure version of [17]. In 2017, Verma et al. [23] proposed an efficient and short CBPS scheme from pairings. In their scheme, the size of the signature was only one elliptic group element (256 bits) and the total computation cost was 53.36 ms. Recently, Verma et al. [25] proposed a CB proxy blind signature scheme. This scheme was also pairing based construction. Thus, in literature several CBPS schemes from pairings were proposed by the research community. However, no CBPS scheme is proposed with pairing free realization.

Due to the development of sensors and actuators, significant deployment of these devices in various applications has emerged. This emerging deployment in industrial applications has inspired the innovative concept of IIoT. However, the dependency on wireless networks has also a big challenge regarding security threats. To discuss these challenges, Sadeghi et al. [19] has proposed several security related issues for IIoT environment. They described that the security in IIoT environment has the same impact as traditional safety. The difference is that the security in IIoT is considered against cyber threats. In IIoT environment, the source shares the data via cloud centric network and thus, the authentication and integrity are important issues. Sadeghi et al. [19] suggested that the attestation of data can solve the issue. Thus, method of digital signing can be a good choice to execute the attestation of data. Recently, several efficient signing methods such as [10,12] has been proposed in the literature. However, the method of delegation of signing power for IIoT is still unavailable.

1.2. Motivation and contribution

As per the discussion by Sadeghi et al. [19], in IIoT environment, method of attestation of data should be used to maintain authentication and integrity. The digital signing of the data can be used to perform the attestation. To deploy on energy trading, Aitzhan and Svetinovic [2] proposed an attestation method by integration of blockchain, multi-signature and anonymous messaging. The construction was based on PKC and thus, suffer from TPQ problem. In 2017, Wei et al. [26] proposed a method of data aggregation for IIoT environment by integrating multi-signature scheme. However, the scheme was ID-based and therefore, suffers from KE and SKD problems. Recently, Jia et al. [10] and Karati et al. [12] have proposed two signature scheme to deploy on IoT and IIoT respectively. Their schemes are based on CLPKC settings and thus, both schemes are subject to SKD problem and key replacement attack. To improve the security in delegation of signing power, Verma et al. [22–24] proposed some efficient and short PS schemes to deploy on wireless sensor networks. However, their schemes use pairing and thus, can be further made more efficient. Thus, the aim of the article is to propose a pairing free proxy signature scheme in certificate-based (PFCBPS) settings to deploy on IIoT environment. The detailed contribution of our article is follows:

- The first CBPS scheme without pairings is proposed. To propose the PFCBPS scheme, the syntax of PFCBPS scheme, network architecture of IIoT environment and attack models are also elaborated in the article.
- The security analysis with respect to unforgeability, integrity, authenticity and various attacks is discussed.
- The efficiency comparison of the proposed PFCBPS scheme is also discussed.

The manuscript roadmap is follows:

The Section 2 presents the basics on ECDLP, network architecture and syntax for the proposed PFCBPS scheme. This section defines the attackers and threat model for the proposed PFCBPS scheme. In Section 3, the proposed PFCBPS scheme is

Table 1
Definitions of symbols used.

Notation	Description
F_q	A prime order (q) finite field
E/F_q	Elliptic curve over F_q
G_T	A cyclic subgroup of E/F_q
1^ν	A security parameter
pk_{CO}	Public key of CO
sk_{CO}	Secret key of CO
ID_J	Identity of John
ID_C	Identity of Charls
pk_{ID_J}	Public key of John
s_{ID_J}	Secret key of John
pk_{ID_C}	Public key of Charls
s_{ID_C}	Secret key of Charls
(R_J, t_J)	Certificate of John
(R_C, t_C)	Certificate of Charls
(w_J, W_J)	Delegation from John to Charls

designed. Section 4 describes the detailed security analysis. In Section 5, performance discussion is proposed. Section 6 concludes the article. The mathematical symbols are defined in Table 1.

2. Preliminaries

This section presents some basics on ECDLP. The network architecture of IIoT environment using delegation, syntax of the proposed PFCBPS scheme. The adversarial oracles and attacks to define provable security in ROM are also presented.

2.1. Basics on elliptic curves

Let, F_q be a finite field with prime q and E/F_q be an elliptic curve over this finite field. Then, E/F_q is defined as the set of all points (x, y) such that $y^2 = x^3 + ax + b$, where a and b are constants from F_q and satisfies $4a^3 + 27b^2 \neq 0$. We consider an additive cyclic subgroup G_T of E/F_q with P as generator, containing all pairs (x, y) of affine co-ordinates on E/F_q and a special point $\mathcal{O}$ at infinity. Since, under addition G_T is a group and so for $s \in \mathbb{Z}_q^*$, sP can be computed as $P + P + P + \ldots + P$ (s times) and is called scalar multiplication.

1. Elliptic Curve Discrete Log Problem (ECDLP) For given random P, $Q \in G_T$, to find $\alpha \in \mathbb{Z}_q^*$ such that $Q = \alpha P$.

2.2. Network architecture of IIoT environment using delegation

In the network architecture of the proposed PFCBPS scheme for IIoT environment, four entities: Cloud Server, CO, Data Owner and End User are involved. The detailed functioning of these is as follows (Fig. 2).

- **Cloud Server:** It works as a medium between other three entities, i.e., CO, Data Owner and End User. It contains public data such as system parameters and pass the signed data from source to receiver.
- **CO:** CO runs the CB-ParGen and generates the system parameters and its secret key. It certifies the ID and public key of each user. Then, it sends system parameters to Cloud Server to be stored.
- **Data Owner:** It is a collection of several industry officials such as system admin, managers, supervisors, etc. along with smart machines (sensors embedded). In the proposed PFCBPS scheme, higher authority works as original signer and its subordinate works as proxy signer. In the case of a smart machine, deploying authority/supervisor works as original signer and sensors as proxy signer. Thus, higher authority (or supervisor) delegates its signing capability to subordinate (or smart machine).
- **End User:** The End User can be a human being or a smart machine (depending on the situation). It receives the proxy signature and verifies this by using public keys of both signers.

In the proposed architecture, CO has sufficient storage and computation facilities. However, Data Owner and End User are smart devices such as mobile phone used by human entity or the sensors embedded. Thus, these smart devices are constrained with respect to both storage and computation power. Thus, in the proposed scheme, the operations executed by these entities must be efficient in terms of computational power and storage. The proposed PFCBPS scheme is designed by keeping these challenges in mind. The scalar multiplication based construction makes it an attractive tool for IIoT environment.

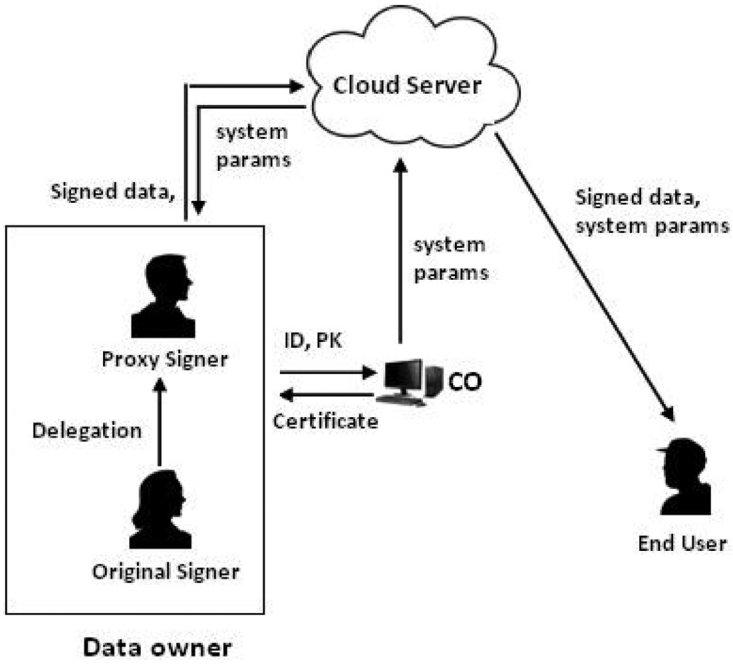


Fig. 2. Network Architecture of IIoT environment using delegation.

2.3. Syntax of the proposed PFCBPS scheme

The proposed PFCBPS scheme is a 7-tuple having polynomial time (PPT) algorithms $(CB-ParGen, CB-KeyGeneration, CB-Certification, CB-Delegation, CB-DelVerification, CB-ProxySigning, CB-PSignVerification)$ with following functionality.

1. *CB-ParGen:* : This PPT algorithm generates system parameter Σ and secret key of CO by taking 1^{ν} (security parameter) as input.
2. *CB-KeyGeneration:* By executing this, each user creates its public, secret key pair. The input is Σ.
3. *CB-Certification:* This interactive PPT algorithm is executed between key owner and certification officer (CO). Key owner gets a certification on its public key and ID from CO after request.
4. *CB-Delegation:* : Original signer runs this PPT algorithm by inputting its secret key, warrant and certificate. The output is delegation to original signer.
5. *CB-DelVerification:* After receiving delegation, proxy signer runs this PPT algorithm and results the validity of delegation. The input is public key of original signer, system parameter Σ and delegation.
6. *CB-ProxySigning:* In this PPT execution, proxy signer outputs a signature on message m. The input is message, secret key of proxy signer, Σ and delegation.
7. *CB-PSignVerification:* : By running this PPT algorithm, end user/verifier outputs accept or reject for the message signature pair. The input is message, signature, Σ, delegation and public keys of both signers.

2.4. Security definitions and attack model of the proposed PFCBPS scheme

1. **Security Attributes:**The security of a proxy signature scheme comprises of the attributes such as Strong Unforgeability, Verifiability, Strong Identifiability, Prevention of Misuse and Strong Undeniability. According to Boldyreva [4,25], the detailed description of these attributes can be defined as follows:
   - Strong Unforgeability: This is the most important and inbuilt security attribute of a signature scheme. It defines that neither a third party (who is not designated as proxy signer) nor an original signer, is capable to sign a document. It is same as forgery of a hand written signatures.
   - Verifiability: If the receiver/verifier is convinced to the delegation agreement between original and proxy signer, then scheme is verifiable.
   - Strong Identifiability: If the proxy signer can easily be identified by an user from the proxy signature, scheme is strong identifiable.
   - Prevention of Misuse: The delegated power cannot be used in any manner other than proxy signing. If the delegated power is used in an illegal manner, the proxy signer is explicitly responsible.
   - Strong Undeniability: A proxy signer cannot deny its functioning on a resulting proxy signature.

2. **Attackers/Adversaries:** With the inspiration from [23], we consider two type of attackers. These attackers are denoted by $\mathcal{F}_1$ and $\mathcal{F}_2$. The power of attackers is defined as follows:
   - Attacker $\mathcal{F}_1$: It has the publicly available data such as Σ and public keys of both signers. Besides these, it has secret keys of both CO and original signer.
   - Attacker $\mathcal{F}_2$: It has the publicly available data such as Σ and public keys of both signers. Besides these, it has secret keys of both CO and proxy signer.

   Therefore, from the above definition $\mathcal{F}_1$ controls both CO and original signer and $\mathcal{F}_2$ controls both CO and proxy signer. Thus, these are the strongest form of adversary for the proposed PFCBPS scheme.
3. **Oracles:** Corresponding to the provable security, we define the following oracles. These oracles are queried by the attackers.
   (a) CB-ParGen(.)-Oracle: Challenger $\mathcal{CH}$ runs the oracle and obtains Σ and secret key of CO.
   (b) CB-KeyGeneration(.)-Oracle: $\mathcal{CH}$ runs this for a requested user ID_i and responds the output to attacker.
   (c) CB-PubKeyReplace(.)-Oracle: Forger runs the oracle and replaces the public key of user ID_i. However, corresponding private key is not known to forger. Forger can run it repeatedly.
   (d) CB-Corruption(.)-Oracle: Forger can request a secret key on user ID_i. $\mathcal{CH}$ checks the results from KeyGen(.)-Oracles and picks the corresponding output. Sends the secret key to forger.
   (e) CB-Certification(.)-oracle: $\mathcal{CH}$ runs the oracle for a user ID_i and sends the results to forger.
   (f) CB-Delegate(.)-Oracle: $\mathcal{CH}$ runs this for a requested user ID_i and warrant w and responds the results to attacker.
   (g) CB-ProxySigning(.)-Oracle: $\mathcal{CH}$ runs the oracle for a requested user (ID_i, w, m) and sends the results to attacker as response.

2.5. Games against $\mathcal{F}_1$ and $\mathcal{F}_2$

The following games are defined towards the provable security of the proposed PFCBPS scheme.

1. **(Security against $\mathcal{F}_1$)Game-1**
   - $\mathcal{CH}$ runs the CB-ParGen(.)-oracle and sends output to forger $\mathcal{F}_1$. $\mathcal{F}_1$ requests CB-KeyGeneration(.), CB-PublicKeyReplace(.), CB-Corruption(.) and CB-ProxySigning(.) oracles and get the responses. It has secret keys of both CO and original signer, so it can run to CB-Certification(.) and CB-Delegate(.) oracles by itself.
   - Output: At last, the following forgeries are displayed by $\mathcal{F}_1$.
     (a) Valid delegation (ID^*, w^*, W^*), while ID^* did not appear during CB-Corruption(.)-Oracle requests and w^* did not appear during CB-Delegate(.)-Oracle requests.
     (b) Valid signature $(ID_J^*, ID_C^*, \sigma^*, m^*)$, (J as original signer and C as proxy signer) while identities ID_J^* and ID_C^* were not queried to CB-Corruption(.)-Oracle and m^* was not queried to ProxySigning(.)-Oracle.

   The success probability is $PrSucc^{EUF}_{\mathcal{F}_1,PFCBPS}$.

   Definition-1: The proposed PFCBPS scheme is EUF-ACMIA, if $PrSucc^{EUF}_{\mathcal{F}_1,PFCBPS}$ is negligible.
2. **(Security against $\mathcal{F}_2$)Game-2**
   - $\mathcal{CH}$ runs the CB-ParGen(.)-oracle and sends output to forger $\mathcal{F}_2$. $\mathcal{F}_2$ requests CB-KeyGeneration(.), CB-Delegate(.), CB-PublicKeyReplace(.) and CB-Corruption(.) oracles and get the responses. It has secret keys of both CO and proxy signer, so it can run to CB-Certification(.) and CB-ProxySigning(.) oracles by itself.
   - Output: At last, the following forgeries are displayed by $\mathcal{F}_2$.
     (a) Valid delegation (ID^*, w^*, W^*), while ID^* did not appear during CB-Corruption(.)-Oracle requests and w^* did not appear during CB-Delegate(.)-Oracle requests.
     (b) Valid signature $(ID_J^*, ID_C^*, \sigma^*, m^*)$, (J as original signer and C as proxy signer) while identity ID_C^* was not queried to CB-Corruption(.)-Oracle and m^* was not queried to ProxySigning(.)-Oracle.

   The success probability is $PrSucc^{EUF}_{\mathcal{F}_2,PFCBPS}$.

   Definition-2: The proposed PFCBPS scheme is EUF-ACMIA, if $PrSucc^{EUF}_{\mathcal{F}_2,PFCBPS}$ is negligible.

3. Proposed PFCBPS scheme

The following are the steps of the proposed PFCBPS scheme.

1. CB-ParGen: By input 1^λ (the security parameter), CO runs the PPT algorithm and generates the system parameter $\Sigma = (E/F_p, G_T, P, p, pk_{CO}, H_0, H_1, H_2)$ and private key sk_{CO} of itself in the following manner:
   - Chooses G_T the cyclic subgroup of points from E/F_q.
   - Randomly selects a secret $s \in \mathbb{Z}_p^*$ and public $P \in G_T$. Then puts $pk_{CO} = sP$ as its public key and $sk_{CO} = s$ its private key.
   - $H_0 : \{0,1\}^* \times G_T \to \mathbb{Z}_p^*, H_1 : \{0,1\}^* \times \{0,1\}^* \times G_T \to \mathbb{Z}_p^*$ and $H_2 : \{0,1\}^* \times \{0,1\}^* \times \{0,1\}^* \times G_T \times G_T \to \mathbb{Z}_p^*$ are three simple cryptographic hash functions.
2. CB-KeyGeneration: Each user U_{ID_i} ($i \in$ {*John, Charls*}) randomly picks $s_{ID_i} \in \mathbb{Z}_p^*$ as her/his secrete key. Then computes $pk_{ID_i} = s_{ID_i}P$ (public key).

User	CO
CB-KeyGeneration(.):	
1. User picks $s_{ID_i} \in_R \mathbb{Z}_p^*$	
2. Computes $pk_{ID_i} = s_{ID_i}P$	
$\xrightarrow{(ID_i, pk_{ID_i})}$	
	CB-Certification(.):
	1. CO picks $r_i \in_R \mathbb{Z}_p^*$ and
	2. Sets $R_i = r_iP$,
	$t_i = r_i + sH_0(ID_i, pk_{ID_i})$
$\xleftarrow{(R_i, t_i)}$	
User checks the validity by	
$t_iP = R_i + H_0(ID_i, pk_{ID_i})pk_{CO}$	

Fig. 3. Certification of ID and public key.

John (Original Signer)	Charls (Proxy Signer)
CB-Delegation(.):	
1. John picks $k_J \in_R \mathbb{Z}_p^*$.	
2. Computes $w_J = R_J + k_JP$	
3. $h_1 = H_1(w, ID_J, pk_{ID_J})$	
$\xrightarrow{(w, w_J, W_J)}$	
	CB-DelVerification(.):
	Charls computes
	1. $h_{0J} = H_0(ID_J, pk_{ID_J})$ and
	2. $h_1 = H_1(w, ID_J, pk_{ID_J})$
	Checks the equality
	$W_JP = w_J + h_{0J}pk_{CO} + h_1pk_{ID_J}$

Fig. 4. Delegation and verification.

3. CB-Certification: User U_{ID_i} requests certification on (ID_i, pk_{ID_i}) from CO. CO generates the certificates by doing the following steps (Fig. 3):
   - CO randomly picks $r_i \in \mathbb{Z}_p^*$.
   - Then, CO computes $R_i = r_iP$ and $t_i = r_i + sH_0(ID_i, pk_{ID_i})$.

   Then, sends (R_i, t_i) to U_{ID_i} as certificate. To verify the certificate, U_{ID_i} checks the equality $t_iP = R_i + H_0(ID_i, pk_{ID_i})pk_{CO}$.
4. CB-Delegation: The original signer, i.e., executive officer (John) computes $w_J = R_J + k_JP$ and $W_J = t_J + k_J + s_{ID_J}H_1(w, ID_J, pk_{ID_J})$ for a randomly chosen $k_J \in \mathbb{Z}_p^*$ and warrant w. Sends (w_J, W_J) to subordinate (proxy signer) Charls as delegation on warrant w (Fig. 4).
5. CB-DelVerification: Charls can check the validity of delegation by checking the equality $W_JP = w_J + H_0(ID_J, pk_{ID_J})pk_{CO} + H_1(w, ID_J, pk_{ID_J})pk_{ID_J}$.
6. CB-ProxySigning: Charls does the following steps (Fig. 5):
   - Selects $k_C \in_R \mathbb{Z}_p^*$ and computes $U_C = w_J + R_C + k_CP$.
   - Computes $\sigma = W_J + k_C + t_C + H_2(w, m, ID_C, pk_{ID_C}, U_C)s_{ID_C}$.

   Then sends (σ, U_C) to receiver as proxy signature on $m \in \{0, 1\}^*$.
7. CB-PSignVerification: The receiver accepts the proxy signature as valid by checking the equality $\sigma P = U_C + (H_0(ID_J, pk_{ID_J}) + H_0(ID_C, pk_{ID_C}))pk_{CO} + H_1(w, ID_J, pk_{ID_J})pk_{ID_J} + H_2(w, m, ID_C, pk_{ID_C}, U_C)pk_{ID_C}$.

Charls (Proxy Signer)		Receiver
CB-ProxySigning(.):		
1. Charls picks $k_C \in_R \mathbb{Z}_p^*$.		
2. Computes $U_C = w_J + R_C + k_C P$		
3. $h_2 = H_1(w, m, ID_C, pk_{ID_C}, U_C)$		
4. $\sigma = W_J + k_C + t_C + h_2 s_{ID_C}$		
	$\xrightarrow{(\sigma, U_C)}$	
		CB-PSignVerification(.):
		Receiver computes
		1. $h_{0J} = H_0(ID_J, pk_{ID_J})$
		2. $h_{0C} = H_0(ID_C, pk_{ID_C})$
		3. $h_1 = H_1(w, ID_J, pk_{ID_J})$
		4. $h_2 = H_1(w, m, ID_C, pk_{ID_C}, U_C)$
		Checks the equality
		$\sigma P = U_C + (h_{0J} + h_{0C}) pk_{CO} + h_1 pk_{ID_J} + h_2 pk_{ID_C}$

Fig. 5. Proxy Sign and verification.

4. Analysis of the proposed PFCBPS scheme

4.1. Proof of correctness

Theorem 1. *The introduced PFCBPS scheme is correct.*

Proof. We know that, $\sigma P = (W_J + k_C + t_C + H_2(w, m, ID_C, pk_{ID_C}, U_C) s_{ID_C})P = R_J + k_J P + H_0(ID_J, pk_{ID_J}) pk_{CO} + H_1(w, ID_J, pk_{ID_J}) pk_{ID_J} + k_C P + R_C + H_0(ID_C, pk_{ID_C}) pk_{CO} = U_C + (H_0(ID_J, pk_{ID_J}) + H_0(ID_C, pk_{ID_C})) pk_{CO} + H_1(w, ID_J, pk_{ID_J}) pk_{ID_J} + H_2(w, m, ID_C, pk_{ID_C}, U_C) pk_{ID_C}$. Thus, verification is consistent. □

4.2. Unforgeability analysis

Theorem 2. *If an attacker $\mathcal{F}_1$ (Type-1) can forge the proposed PFCBPS scheme in ROM with probability of success PrSucc $\geq \epsilon$, within time bound t. Then, a PPT $\mathcal{B}$ can be constructed to solve the random instance of ECDLP with probability of success $PrSucc_{\mathcal{B}, G_T}^{(ECDLP)} \geq \frac{1}{q_{kg}} \left(1 - \frac{1}{q_s}\right)^{q_{kg}} \epsilon$. Where, q_{kg} and q_s be the number of requests asked by $\mathcal{F}_1$ during key generation and signing.*

Proof. The challenge is given instance $(P, \alpha P = Q)$ of ECDLP in G_T. $\mathcal{B}$ stores all responses of the requests (made by $\mathcal{F}_1$) in the lists represented by $L_{KG}, L_H^0, L_H^1, L_H^2, L_D, L_C$ and L_{PS}. Initially, all lists are set empty. The list L_{KG} stores the responses from CB-KeyGeneration(.)-Oracle, L_H^0 stores the responses from H_0(.)-Oracle, L_H^1 stores the responses from H_1(.)-Oracle, L_H^2 stores the responses from H_2(.)-Oracle, L_D stores the responses from CB-Delegate(.)-Oracle, L_C stores the responses from CB-Certification(.)-Oracle and L_{PS} stores the responses from CB-ProxySigning(.)-Oracle. □

- CB-ParGen(.)-Oracle: Initially, $\mathcal{B}$ sets $pk_{CO} = sP$ and $pk_{ID_J} = aP$ for randomly chosen $s, a \in \mathbb{Z}_p^*$. Then, $\mathcal{B}$ gives pk_{ID_J}, s, a and $\Sigma = (E/F_p, G_T, P, p, pk_{CO}, H_0, H_1, H_2)$ to $\mathcal{F}_1$. Now, $\mathcal{B}$ plays the following game to output a solution of ECDLP by taking $(P, \alpha P = Q)$ as input.
- CB-KeyGeneration(.)-Oracle: $\mathcal{F}_1$ makes CB-KeyGeneration(.)-Oracle queries on $ID_i (1 \leq i \leq q_{kg})$ to get corresponding keys. $\mathcal{B}$ checks L_{KG} to match the requested ID_i. If the matching tuple $(ID_i, s_{ID_i}, pk_{ID_i})$ is found, sends pk_{ID_i} to $\mathcal{F}_1$. Otherwise, does the following:
  - If $i \neq C$ for some $(1 \leq k \leq q_{kg})$, $\mathcal{B}$ chooses $s_{ID_i} \in \mathbb{Z}_p^*$ and puts $s_{ID_i} P = pk_{ID_i}$ and adds $(ID_i, s_{ID_i}, pk_{ID_i})$ to L_{KG} and pk_{ID_i} is response to $\mathcal{F}_1$.
  - If $i = k$ for some $(1 \leq k \leq q_{kg})$, $\mathcal{B}$ puts $Q = pk_{ID_i}$ and adds $(ID_i, \perp, pk_{ID_i})$ to L_{KG} and pk_{ID_i} is response to $\mathcal{F}_1$.
- $H_0(.) - Oracle$: On requested (pk_{ID_i}, ID_i) to $H_0(.) - Oracle$, $\mathcal{B}$ checks the L_H^0 to match (pk_{ID_i}, ID_i, h_i^0). If tuple exists, picks h_i^0. Otherwise, $\mathcal{B}$ picks $h_i^0 \in \mathbb{Z}_p^*$ in a random fashion and puts $h_i^0 = H_0(ID_i, pk_{ID_i})$. L_H^0 is updated by adding (pk_{ID_i}, ID_i, h_i^0). At last, h_i^0 is response to $\mathcal{F}_1$.

- $H_1(.)-Oracle$: On queried (pk_{ID_i}, ID_i, w_i) to $H_1(.)-Oracle$, $\mathcal{B}$ scans the L_H^1 to match $(pk_{ID_i}, ID_i, w_i, h_i^1)$. If matching exists, picks h_i^1. Otherwise, $\mathcal{B}$ picks $h_i^1 \in \mathbb{Z}_p^*$ in a random fashion and sets $h_i^1 = H_1(w_i, ID_i, pk_{ID_i})$. L_H^1 is updated by adding $(pk_{ID_i}, ID_i, w_i, h_i^1)$. Finally, h_i^1 is forwarded to $\mathcal{F}_1$.
- $H_2(.)-Oracle$: On queried $(pk_{ID_i}, ID_i, w_i, m_i, U_i)$ to $H_2(.)-Oracle$, $\mathcal{B}$ scans the L_H^2 to match $(pk_{ID_i}, ID_i, w_i, m_i, U_i, h_i^2)$. If matching exists, picks h_i^2. Otherwise, $\mathcal{B}$ picks $h_i^2 \in \mathbb{Z}_p^*$ in a random fashion and sets $h_i^2 = H_1(w_i, m_i, ID_i, pk_{ID_i}, U_i)$. L_H^2 is updated by adding $(pk_{ID_i}, ID_i, w_i, m_i, U_i, h_i^2)$. Finally, h_i^2 is forwarded to $\mathcal{F}_1$.
- CB-PublicKeyReplace(.)-Oracle: $\mathcal{F}_1$ controls both CO and John (original signer), it can query (ID_J, pk'_{ID_J}) to replace $(ID_J, s_{ID_J}, pk_{ID_J})$. $\mathcal{B}$ picks the tuple from L_{KG} and replace it with $(ID_J, \perp, pk'_{ID_J})$.
- CB-Corruption(.)-Oracle: On queried ID_i, $\mathcal{B}$ responds in the following manner.
  - If $i \neq C$, $\mathcal{B}$ scans L_{KG} to get the matching tuple $(ID_i, s_{ID_i}, pk_{ID_i})$. If the matching is successful, s_{ID_i} is responded to $\mathcal{F}_1$. Otherwise, picks $s_{ID_i} \in \mathbb{Z}_p^*$ in a random fashion and puts $s_{ID_i}P = pk_{ID_i}$. $\mathcal{B}$ adds $(ID_i, s_{ID_i}, pk_{ID_i})$ to L_{KG} and x_{ID_i} is response to $\mathcal{F}_1$.
  - If $i = C$, $\mathcal{B}$'s response is $\perp$.
- CB-ProxySigning(.)-Oracle: On a requested (m_i, w_i, ID_j, ID_C), where ID_J and ID_C are John's and Charls's identities respectively. $\mathcal{B}$ responds in the following fashion.
  - $\mathcal{B}$ scans all L_H^0, L_H^1 and L_H^2 and obtains the values $H_0(ID_j, pk_{ID_j}), H_0(ID_C, pk_{ID_C}), H_1(w_i, ID_j, pk_{ID_j})$ and $H_2(w_i, m_i, ID_C, pk_{ID_C}, U_C)$.
  - $\mathcal{B}$ runs CB-Certification(.) and CB-Delegate(.) oracles to obtain certificates on $H_0(ID_j, pk_{ID_j})$ and $H_0(ID_C, pk_{ID_C})$ and valid delegation (w_i, W_i).
  - $\mathcal{B}$ scans L_{KG} to obtain x_{ID_C} from the tuple $(ID_C, s_{ID_C}, pk_{ID_C})$. If $s_{ID_C} = \perp$, $\mathcal{B}$ aborts and fails. Otherwise, picks s_{ID_C}. Finally, either $\mathcal{B}$ outputs failure or computes $U_C^i = w_i + R_C + k_C^i P$ and $\sigma_i = W_J^i + k_J^i + t_J + H_2(w_i, m_i, ID_C, pk_{ID_C}, U_C^i)s_{ID_C}$ for a random $k_J^i \in \mathbb{Z}_p^*$.
- Output: Thus, either $\mathcal{F}_1$ fails or outputs the following forgery.
  - (w^*, W^*) is a valid delegation, while $(w^*, W^*, ID_J, pk_{ID_J}) \notin L_D$.
  - (m^*, U^*, σ^*) is a valid signature, while $(m^*, U^*, \sigma^*, ID_J, pk_{ID_J}, ID_C, pk_{ID_C}) \notin L_{PS}$.
- Solution to ECDLP: According to the above game, either $\mathcal{F}_1$ fails or outputs the above forgery. $\mathcal{F}_1$ controls CO and John, so for the signature forgery, by applying Forking lemma [18] $\mathcal{B}$ outputs two valid proxy signatures $(w^*, W^*, m^*, U^*, \sigma^*)$ and $(w^*, W^*, m^*, U^*, \sigma^{**})$ on a same message m^* and warrant w^* with the same random tape but with different hashing values from $H_2(.)$ oracle. Since, both are valid signatures. Thus, $\sigma^* P = U_C^* + (h_{J0}^* + h_{C0}^*)pk_{CO} + h_1^* pk_{ID_J} + h_2^* pk_{ID_C}$ and $\sigma^{**}P = U_C^* + (h_{J0}^* + h_{C0}^*)pk_{CO} + h_1^* pk_{ID_J} + h_2^{**} pk_{ID_C}$. Taking $pk_{ID_C} = Q = \alpha P$, we get $\alpha = \frac{\sigma^* - \sigma^{**}}{h_2^* - h_2^{**}}$, i.e., solution to ECDLP.

Breaching Probability: Based on the above game, the $PrSucc_{\mathcal{B},G_T}^{ECDLP}$ to solve ECDLP can be computed from the following events.

E_1: $\mathcal{B}$ does not fail during game.

E_2: $\mathcal{F}_1$ can successfully create a valid forged signature.

E_3: $\mathcal{F}_1$ played above game to a targeted identity.

Thus, $P[E_1] \geq \left(1 - \frac{1}{q_s}\right)^{q_{kg}}, P[E_2|E_1] \geq \epsilon$ and $P[E_3|E_1 \wedge E_2] = \frac{1}{q_{kg}}$. Therefore, $PrSucc_{\mathcal{B},G_T}^{(ECDLP)} \geq \frac{1}{q_{kg}}\left(1 - \frac{1}{q_s}\right)^{q_{kg}}\epsilon$. Where, q_{kg} and q_s be the number of requests asked by $\mathcal{F}_1$ to CB-KeyGeneration(.) and CB-ProxySigning(.) oracles.

Theorem 3. *If an attacker $\mathcal{F}_2$ (Type-2) can forge the proposed PFCBPS scheme in ROM with probability of success $PrSucc \geq \epsilon$, within time bound t. Then, a PPT $\mathcal{B}$ can be constructed to solve the random instance of ECDLP with probability of success $PrSucc_{\mathcal{B},G_T}^{(ECDLP)} \geq \frac{1}{q_{kg}}\left(1 - \frac{1}{q_w}\right)^{q_{kg}}\epsilon$. Where, q_{kg} and q_w be the number of requests asked by $\mathcal{F}_1$ during key generation and delegation.*

Proof. The challenge is given instance $(P, \alpha P = Q)$ of ECDLP in G_T. $\mathcal{B}$ stores all responses of the requests (made by $\mathcal{F}_1$) in the lists represented by $L_{KG}, L_H^0, L_H^1, L_H^2, L_D, L_C$ and L_{PS}. Initially, all lists are set empty. The list L_{KG} stores the responses from CB-KeyGeneration(.)-Oracle, L_H^0 stores the responses from $H_0(.)$-Oracle, L_H^1 stores the responses from $H_1(.)$-Oracle, L_H^2 stores the responses from $H_2(.)$-Oracle, L_D stores the responses from CB-Delegate(.)-Oracle, L_C stores the responses from CB-Certification(.)-Oracle and L_{PS} stores the responses from CB-ProxySigning(.)-Oracle. □

- CB-ParGen(.)-Oracle: Initially, $\mathcal{B}$ sets $pk_{CO} = sP$ and $pk_{ID_C} = aP$ for randomly chosen $s, a \in \mathbb{Z}_p^*$. Then, $\mathcal{B}$ gives pk_{ID_C}, s, a and $\Sigma = (E/F_p, G_T, P, p, pk_{CO}, H_0, H_1, H_2)$ to $\mathcal{F}_2$. Now, $\mathcal{B}$ plays the following game to output a solution of ECDLP by taking $(P, \alpha P = Q)$ as input.
- CB-KeyGeneration(.)-Oracle: $\mathcal{F}_2$ makes CB-KeyGeneration(.)-Oracle queries on $ID_i (1 \leq i \leq q_{kg})$ to get corresponding keys. $\mathcal{B}$ checks L_{KG} to match the requested ID_i. If the matching tuple $(ID_i, s_{ID_i}, pk_{ID_i})$ is found, sends pk_{ID_i} to $\mathcal{F}_1$. Otherwise, does the following:
  - If $i \neq k$ for some $(1 \leq k \leq q_{kg})$, $\mathcal{B}$ chooses $s_{ID_i} \in \mathbb{Z}_p^*$ and puts $s_{ID_i}P = pk_{ID_i}$ and adds $(ID_i, s_{ID_i}, pk_{ID_i})$ to L_{KG} and pk_{ID_i} is response to $\mathcal{F}_1$.
  - If $i = k$ for some $(1 \leq k \leq q_{kg})$, $\mathcal{B}$ puts $Q = pk_{ID_i}$ and adds $(ID_i, \perp, pk_{ID_i})$ to L_{KG} and pk_{ID_i} is response to $\mathcal{F}_1$.
- $H_0(.)$, $H_1(.)$ and $H_2(.)$ oracles are executed in the same fashion as in Theorem 2. The outputs are also same.

- CB-PublicKeyReplace(.)-Oracle: $\mathcal{F}_2$ controls both CO and Charls (proxy signer), it can query (ID_C, pk'_{ID_C}) to replace $(ID_C, s_{ID_C}, pk_{ID_C})$. $\mathcal{B}$ picks the tuple from L_{KG} and replace it with $(ID_C, \perp, pk'_{ID_C})$.
- CB-Corruption(.)-Oracle: On queried ID_i, $\mathcal{B}$ responds in the following manner.
  - If $i \neq J$, $\mathcal{B}$ scans L_{KG} to get the matching tuple $(ID_i, s_{ID_i}, pk_{ID_i})$. If the matching is successful, s_{ID_i} is responded to $\mathcal{F}_2$. Otherwise, picks $s_{ID_i} \in \mathbb{Z}_p^*$ in a random fashion and puts $s_{ID_i}P = pk_{ID_i}$. $\mathcal{B}$ adds $(ID_i, s_{ID_i}, pk_{ID_i})$ to L_{KG} and s_{ID_i} is response to $\mathcal{F}_2$.
  - If $i = J$, $\mathcal{B}$'s response is $\perp$.
- CB-Delegate(.)-Oracle: On a requested (w_i, ID_i), $\mathcal{B}$ responds in the following fashion.
  - $\mathcal{B}$ runs CB-Certification(.)-Oracle and obtain the corresponding certificates of (ID_i).
  - $\mathcal{B}$ scans L_{KG} to match the tuple $(ID_i, s_{ID_i}, pk_{ID_i})$. If matching is done, sets $W_i = t_i + k_i + s_{ID_i}h_i^1$, where t_i and h_i^1 are obtained from L_C and L_H^1 lists and k_i is random from $\mathbb{Z}_p^*$. Otherwise, reports $\perp$.

  In either case, L_D is updated.
- CB-ProxySigning(.)-Oracle: On a requested (m_i, w_i, ID_J, ID_C), where ID_J and ID_C are John's and Charls's identities respectively. $\mathcal{B}$ responds in the following fashion.
  - $\mathcal{B}$ scans all L_H^0, L_H^1 and L_H^2 and obtains the values $H_0(ID_j, pk_{ID_j}), H_0(ID_C, pk_{ID_C}), H_1(w_i, ID_j, pk_{ID_j})$ and $H_2(w_i, m_i, ID_C, pk_{ID_C}, U_C)$.
  - $\mathcal{B}$ runs CB-Certification(.) oracle to obtain certificates on $H_0(ID_j, pk_{ID_j})$ and $H_0(ID_C, pk_{ID_C})$.
  - Corresponding delegation is obtained from L_D.
  - Finally, either $\mathcal{B}$ outputs failure or computes $U_C^i = w_i + R_C + k_C^i P$ and $\sigma_i = W_J^i + k_J^i + t_J + H_2(w_i, m_i, ID_C, pk_{ID_C}, U_C^i)s_{ID_C}$ for a random $k_J^i \in \mathbb{Z}_p^*$.
- Output: Thus, either $\mathcal{F}2$ fails or outputs the following forgery.
  - (w^*, W^*) is a valid delegation, while $(w^*, W^*, ID_J, pk_{ID_J}) \notin L_D$.
  - (m^*, U^*, σ^*) is a valid signature, while $(m^*, U^*, \sigma^*, ID_J, pk_{ID_J}, ID_C, pk_{ID_C}) \notin L_{PS}$.
- Solution to ECDLP: According to the above game, either $\mathcal{F}_2$ fails or outputs the above forgery. $\mathcal{F}_2$ controls CO and Charls, so for the delegation forgery, by applying Forking lemma [18] $\mathcal{B}$ outputs two valid delegations (w_J^*, W_J^*) and (w_J^*, W_J^{**}) on a same warrant w^* with the same random tape but with different hashing values from $H_1(.)$ oracle. Since, both are valid delegations. Thus, $W_J^*P = R_J + k_JP + h_0^* pk_{CO} + h_1^* pk_{ID_J}$ and $W_J^{**}P = R_J + k_JP + h_0^* pk_{CO} + h_1^{**} pk_{ID_J}$ Taking $pk_{ID_J} = Q = \alpha P$, we get $\alpha = \frac{W_J^* - W_J^{**}}{h_1^* - h_1^{**}}$, i.e., solution to ECDLP.

Breaching Probability: Based on the above game, the $PrSucc_{\mathcal{B},G_T}^{ECDLP}$ to solve ECDLP can be derived from the following events.

E_1: $\mathcal{B}$ does not fail during game.

E_2: $\mathcal{F}_2$ can successfully create a valid forgery.

E_3: $\mathcal{F}_2$ played above game to a targeted identity.

Thus, $P[E_1] \geq \left(1 - \frac{1}{q_w}\right)^{q_{kg}}$, $P[E_2|E_1] \geq \epsilon$ and $P[E_3|E_1 \wedge E_2] = \frac{1}{q_{kg}}$. Therefore, $PrSucc_{\mathcal{B},G_T}^{(ECDLP)} \geq \frac{1}{q_{kg}}\left(1 - \frac{1}{q_w}\right)^{q_{kg}}\epsilon$. Where, q_{kg} and q_w be the number of requests asked by $\mathcal{F}_2$ during key generation and delegation.

4.3. Security analysis

- Strong Unforgeability: In the formal security analysis, we have considered two strongest forms of attackers. Thus, to forge the proposed PFCBPS scheme, an attacker needs to solve ECDLP. Therefore, the scheme presents Strong Unforgeability under the intractability of ECDLP. No third party (who is not designated as proxy signer) can forge the proposed PFCBPS scheme.
- Verifiability: From the proxy sign verification phase of the proposed PFCBPS scheme, it is clear that verification needs the identities and public keys of both signers. The original signerâs public key (pk_{ID_J}) is used with hashing of warrant and proxy signerâs public key is used with hashing of message. The warrant contains the information on message scope, validity, etc. Thus, lack of any one will display reject. Thus, verifier is convinced to the delegation agreement, i.e., scheme is verifiable.
- Strong Identifiability: The discussion of verifiability shows that the public keys of both signers are used with specific hashing values. Therefore, after seeing verification phase, everyone can identify both signers. Thus, the proposed scheme is Strong identifiable.
- Prevention of Misuse: The proof of formal security shows that the malicious proxy signer cannot generate a proxy signature in dispute. So, we can conclude that the delegation cannot be misused by proxy signer.
- Strong Undeniability: From the formal security analysis, Theorems 2 and 3 show that no third party other than proxy signer is capable to generate a valid proxy signature. The verification process also shows that public keys of CO, original signer and proxy signer are needed to verify the signature. In detail, certificates are authenticated by CO, warrant by original signer and message by proxy signer. Therefore, proxy signer cannot deny its behavior on resulting proxy signature. Thus, scheme presents Strong undeniability.
- Man-in-the-middle (MITM) attack: During the attack, an unauthorized person or device tries to capture the data between source and the sender. Then, it can result the loss of originality or freshness. In the proposed PFCBPS scheme,

Table 2
Computational costs of different operations.

Operation	Notation	Cost (ms)
Bilinear Pairing	t_e	20.01
Scalar Multiplication	t_{s_m}	6.67
Modular Inversion	t_{Inv}	2.67
Map to point hash	t_h	6.67

Table 3
Performance comparison of the proposed PFCBPS scheme with exiting competitive schemes.

Scheme	DelGen	ms	DelVer	ms	ProxySign	ms	ProxySVer	ms
PFCBPS	$1t_{s_m}$	6.67	$3t_{s_m}$	20.01	$1t_{s_m}$	6.67	$2t_{s_m}$	13.34
[8]	$1t_{s_m}$	6.67	$4t_{s_m}$	26.68	$1t_{s_m} + 1t_{inv}$	9.34	$7t_{s_m}$	46.69
[5]	$1t_{s_m}$	6.67	$3t_{s_m}$	20.01	$1t_{s_m} + 1t_{inv}$	9.34	$6t_{s_m}$	40.02
[20]	$2t_{s_m} + 1t_h$	20.01	$1t_{s_m} + 1t_h + 3t_e$	73.37	$2t_{s_m} + 1t_h$	20.01	$2t_{s_m} + 2t_h + 3t_e$	86.71
[9]	$1t_{s_m} + 1t_h$	13.34	$1t_h + 2t_e$	46.69	$2t_{s_m} + 1t_h$	20.01	$2t_{s_m} + 2t_h + 3t_e$	86.71
[23]	$1t_{s_m} + 1t_h$	13.34	$1t_h + 2t_e$	46.69	$1t_{s_m} + 1t_{inv}$	13.34	$2t_e + 1t_{s_m} + 1t_h$	53.36

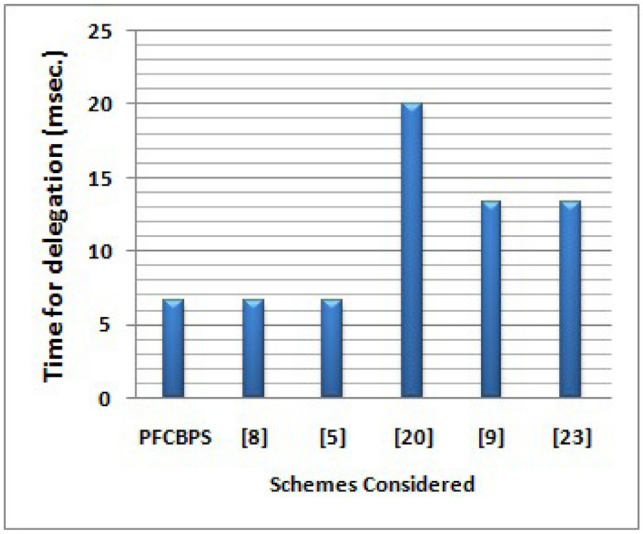


Fig. 6. Computational cost comparison of delegation.

Theorems 1–3 prove the correctness and unforgeability. Thus, any alteration to the message will output the rejection of signature during verification. The unforgeability and warrant protect the freshness as attacker is unable to produce a forgery on old messages. Therefore, MITM attack is unsuccessful in the proposed scenario.

5. Performance discussion

5.1. Performance and length comparison

To discuss the performance comparison, the symbols and computational costs of the different operations are given in Table 2 [13].

In Table 3, we have considered the four phases (Delegation, Delegation Verify, Proxy Signing and Proxy Sign verify) of the proposed PFCBPS scheme. The reason of calculating computational costs separately is that different phases are executed by different entities (along with their constrained). Thus, we found that the delegation phase of the proposed PFCBPS scheme is executed in 6.67 ms which is either same or less than the schemes considered [5,8,9,20,23] (Fig. 6). The delegation verify phase of the proposed PFCBPS scheme takes 20.01 ms which is 100% of [5] (i.e. equal to [5]), 75% of [8], 27.27% of [20], 42.86% of both [9,23]. Therefore, this phase of the proposed PFCBPS scheme is efficient (Fig. 7). In the same manner, the proxy signing phase of the proposed PFCBPS scheme takes 6.67 ms and it is 71.41% of [5,8], 50% of [23] and 33.33% of [9,20]. Thus, signing phase of the proposed PFCBPS scheme is more efficient than other schemes (Fig. 8). The proxy sign verify phase of the proposed PFCBPS scheme is running in 13.34 ms which is 28.57% of [8], 33.33% of [5], 15.39% of [9,20] and 25% of [23] (Fig. 9). Therefore, each phase of the proposed PFCBPS scheme is efficient (Fig. 10). It is also evident from the comparison that in a cumulative manner, the proposed scheme is the most efficient scheme.

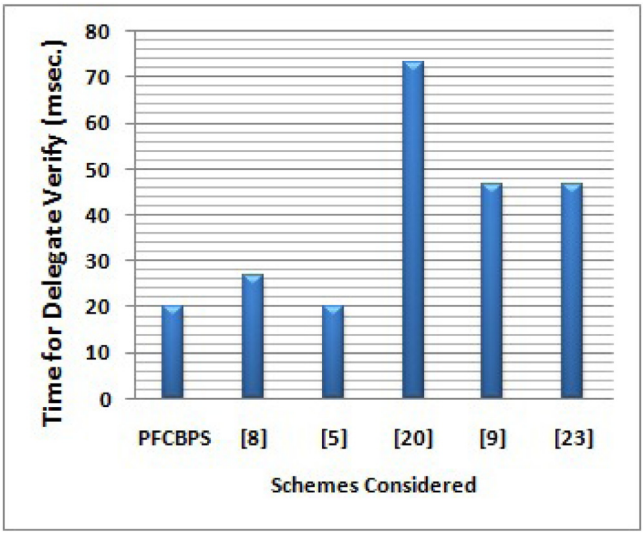


Fig. 7. Computational cost comparison of delegation verify.

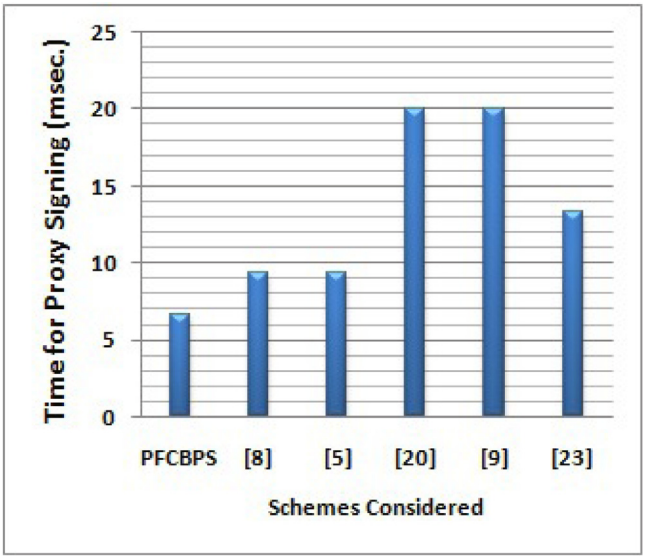


Fig. 8. Computational cost comparison of signing.

The lengths (in bits) of delegated information and signature are also considered. The elements from G_T and $\mathbb{Z}_p^*$ have 256 bit length [1]. Thus, from Table 4, each of the delegation and signature length of the proposed PFCBPS scheme are 512 (bits). Therefore, the scheme by Verma et al. [23] is shorter in terms of both delegation and signature lengths. However, it is pairing-based construction and therefore, inefficient. The scheme by Hu et al. [9] has shorter delegation length, however, it is also a pairing-based scheme, therefore, is also inefficient. While, other schemes such as [5,8,20] have larger signature as well as delegation lengths along with less efficiency in terms of computational cost (Figs. 11 and 12).

5.2. Appealing to IIoT

Now, we discuss that why the proposed scheme is the most appealing to IIoT environment. In general, most of the executives and their subordinates use mobile phone to manage their official communications. In different machine based environment, the deployed sensors also disseminate collected data. Thus, all four phases (Delegation, Delegation Verify, Proxy Signing and Proxy Sign verify) are executed on smart devices having limited power and storage. As per the above discussion, the different phases, i.e., Delegation, Delegation Verify, Proxy Signing and Proxy Sign verify of the proposed schemes take 6.67 ms, 20.01 ms, 6.67 ms and 13.34 ms respectively. Thus, the computation costs can easily be managed by

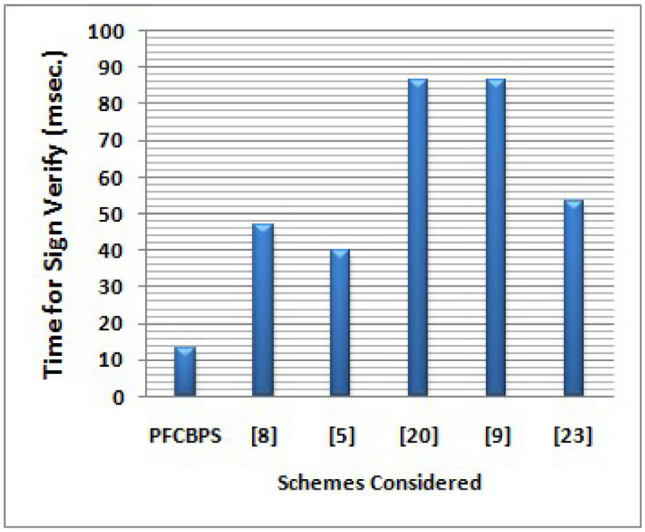


Fig. 9. Computational cost comparison of sign verify.

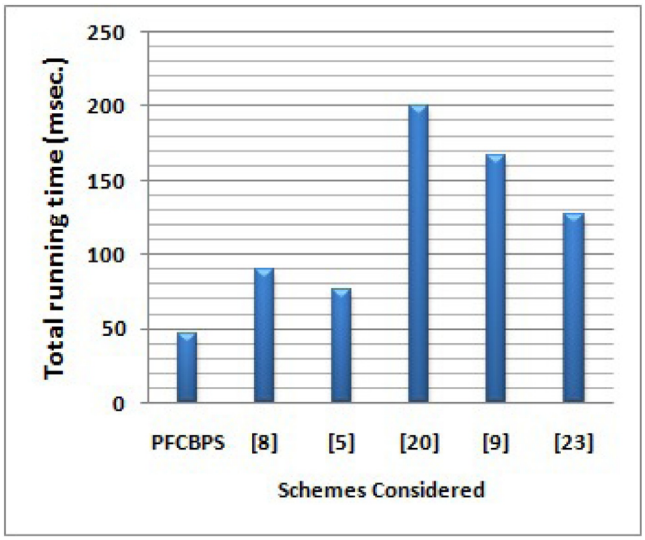


Fig. 10. Total computational cost comparison.

Table 4
Bit length comparison of the proposed PFCBPS scheme with exiting competitive schemes.

Scheme	Delegation	bits	Signature	bits
PFCBPS	$1\|G_t\|+1\|\mathcal{Z}_p\|$	512	$1\|G_t\|+1\|\mathcal{Z}_p\|$	512
[8]	$2\|G_t\|+1\|\mathcal{Z}_p\|$	768	$4\|G_t\|+\|\mathcal{Z}_p\|$	1280
[5]	$2\|G_t\|+1\|\mathcal{Z}_p\|$	768	$4\|G_t\|+2\|\mathcal{Z}_p\|$	1536
[20]	$2\|G_t\|$	512	$3\|G_t\|$	768
[9]	$1\|G_t\|$	256	$2\|G_t\|$	512
[23]	$1\|G_t\|$	256	$1\|G_t\|$	256

any smart device having limited resources. The size of delegation and signature is also 512 bits only, which is also bearable to any wireless network between such smart devices.

To discuss the implementation in detail, we consider an example of a machine deployed with sensors. In this system, the system administrator works as CO, the supervisor of the machine is original signer and the sensors are proxy signer. The supervisor delegates its signing capacity to sensor and the certificates and system parameters are also embedded on the sensor. After collecting data, sensor uses proxy signing method (which costs 6.67 ms) to attest the data before communica-

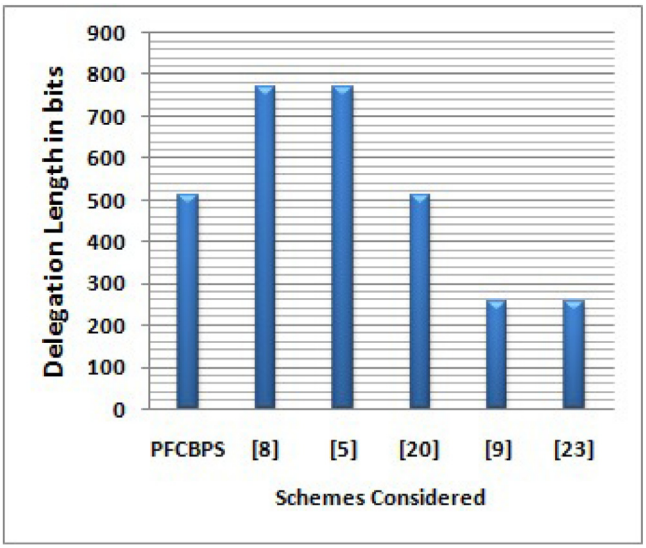


Fig. 11. Bit-length Comparison of Delegation.

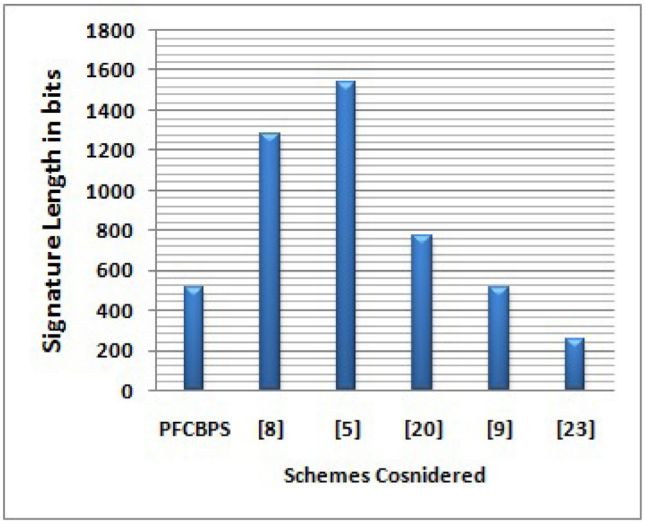


Fig. 12. Bit-length Comparison of signature.

tion. The receiver end may be a mobile phone, computer or another sensor. Then receiver runs verification process (which costs 13.34 ms). Thus, due to the least computational cost, every phase of the proposed scheme is bearable to any smart device deployed in IIoT environment. Therefore, the proposed PFCBPS scheme is the most appealing attestation method in IIoT environment.

6. Conclusion

The emerging revolution in industrial environment inspired the replacement of logic based units with cyber-physical systems. In industrial environment, the deployment of such cyber-physical systems has been shaped due to IoT. However, the need of authentication has also become more important. To provide an efficient authentication mechanism, this article proposed a certificate-based proxy signature (PFCBPS) scheme. The pairing free construction of the proposed PFCBPS scheme has made it the most efficient authentication method. The performance comparison has also proved this efficiency fact. The security analysis (formal and informal) has also been proposed to show that the PFCBPS scheme is the most appealing to IIoT environment with respect to both attributes, i.e., efficiency and security.

Recently, the distributed ledger systems are becoming popular. However, perhaps the proposed scheme may support the distributed ledger. Thus, to implement the scheme in distributed ledger systems further research should be done on the line. In future, the blockchain based authentication method for IIoT environment by using delegation will be proposed.

Declaration of Competing Interest

We have no declaration of interest.

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